Research Question: How effectively can a parsimonious ARIMA model capture the temporal dependence in U.S. industrial production, and how sensitive are its 50-month prediction intervals to assumptions about the innovation distribution?

Source: [Industrial Production: Total Index (INDPRO) | FRED | St. Louis Fed]

Data Source and Motivation

The Industrial Production Index (INDPRO), available through the Federal Reserve Bank of St. Louis's FRED database, measures U.S. industrial production activity.

We analyze 240 monthly observations of the seasonally adjusted index (2017 = 100), covering October 2006 through September 2026. The data were downloaded on October 22, 2026. This 20-year window captures recent industrial production dynamics, including the 2008 financial crisis and the COVID-19 pandemic, while providing sufficient observations for ARIMA modeling.

These major disruptions also provide a useful setting for evaluating how conventional ARIMA models handle large economic shocks.

Exploratory Data Analysis and Model Identification

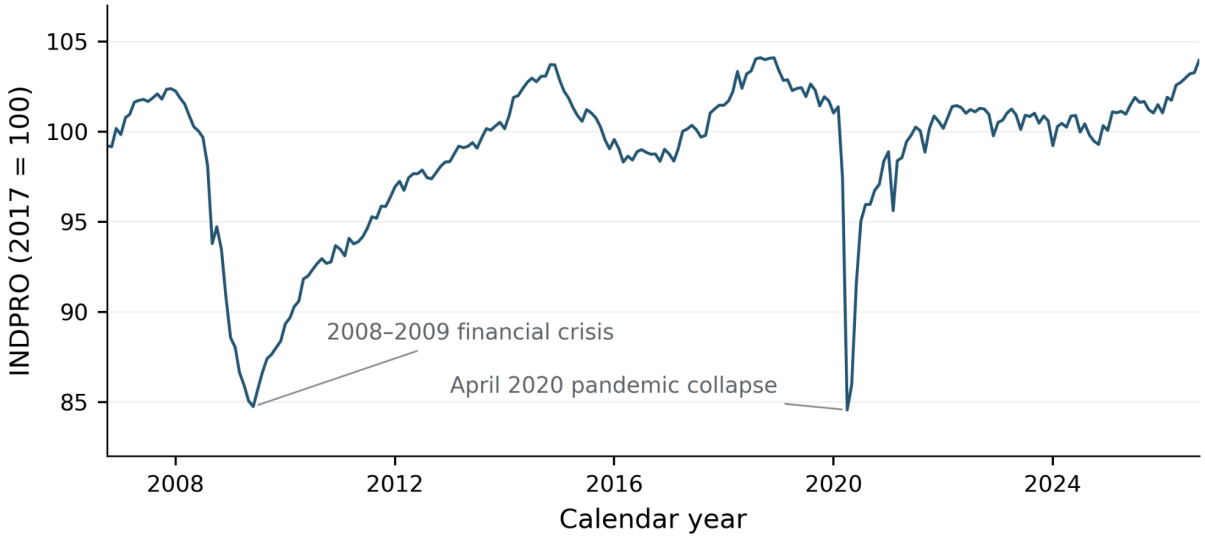


Figure 1. U.S. Industrial Production Index, October 2006–September 2026.

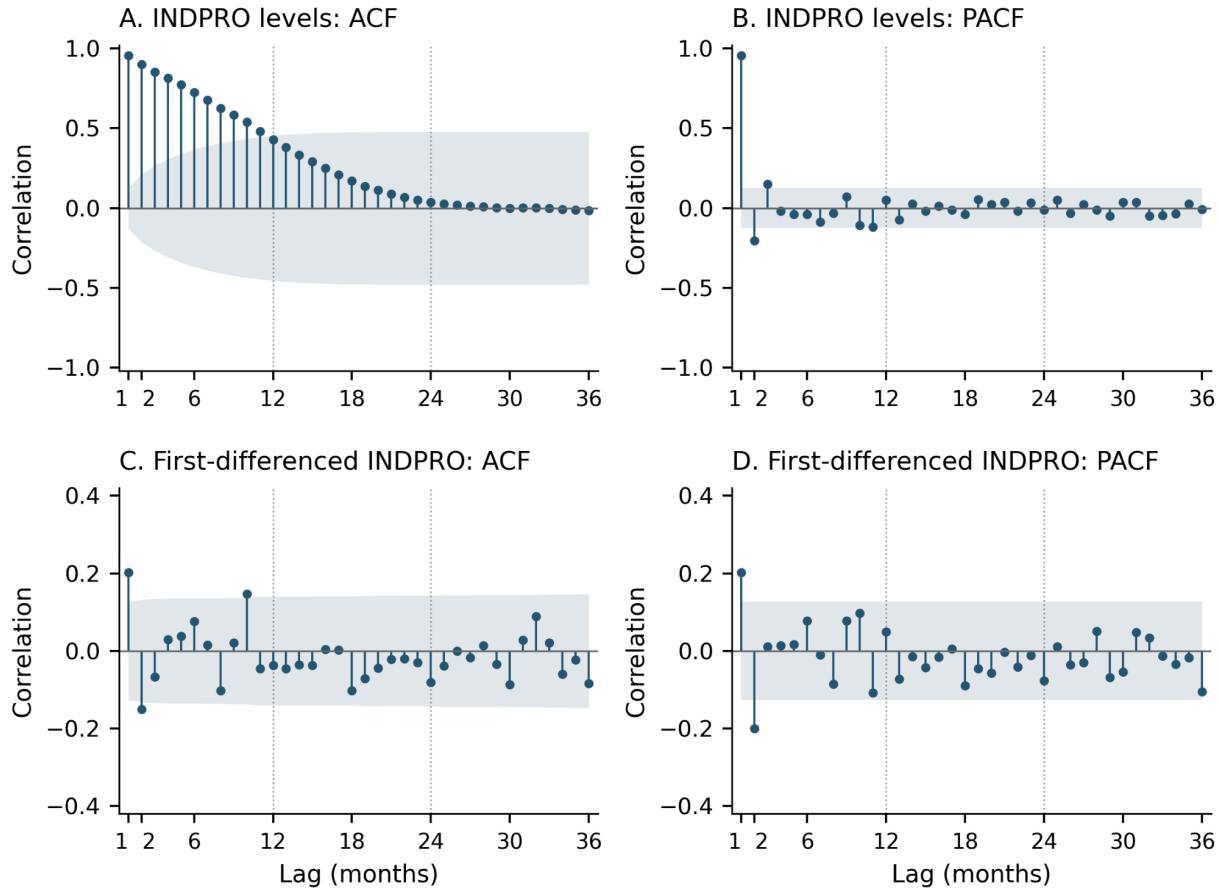


Figure 2. ACF and PACF of INDPRO before and after first differencing.

95% pointwise bands: Bartlett ACF; normal-approximation Yule–Walker PACF.
Levels: n = 240; differences: n = 239. Vertical scales differ by row; lag 0 omitted.

Figure 1 shows sharp declines during the 2008–2009 financial crisis and the COVID-19 pandemic, followed by a gradual recovery after the former and a rapid rebound after the latter. Although these recoveries suggest some reversal of large shocks, they do not establish mean reversion toward a fixed long-run level. The level ACF decays slowly from 0.9576 at lag 1 (Figure 2A), indicating strong persistence and possible nonstationarity. After first differencing, the lag-1 ACF falls to 0.2024, with most subsequent correlations relatively small (Figure 2C). We therefore set d=1, while recognizing that this specification assumes persistent effects of shocks on the series level.

The differenced PACF has prominent spikes at lags 1 and 2 (Figure 2D), suggesting low-order autoregressive dependence. We therefore consider ARIMA(1,1,0) and ARIMA(2,1,0), alongside ARIMA(0,1,1), ARIMA(0,1,2), and ARIMA(1,1,1) to allow for moving-average effects. ARIMA(0,1,0) serves as a random-walk benchmark. An isolated ACF spike near lag 10 does not alone justify a high-order term.

Beyond first differencing, no further transformation appears necessary. A log transformation is not warranted because fluctuations do not systematically increase with the series level; the largest changes instead coincide with major economic shocks. There is also no clear annual seasonal pattern: INDPRO is already seasonally adjusted, and the differenced ACF shows no pronounced spikes at lags 12 or 24. We therefore retain the original index scale and apply no additional seasonal differencing before comparing ARIMA specifications using AICc and residual diagnostics.

Model Selection and Residual Diagnostics

Order	AICc, no drift	AICc, drift
(0,1,0)	767.071227	769.039639
(1,1,0)	759.115714	761.120517
(2,1,0)	751.485410	753.494224
(0,1,1)	754.319642	756.327361
(0,1,2)	753.111109	755.126091
(1,1,1)	754.494171	756.515542

Table 1. *AICc Comparison of Candidate ARIMA Models*

Table 1 compares candidate ARIMA models using AICc, which penalizes model complexity. ARIMA(2,1,0) without drift has the lowest AICc (751.485), followed by ARIMA(0,1,2) without drift (753.111). The small difference (1.626) indicates that both remain plausible. Adding drift to ARIMA(2,1,0) raises its AICc to 753.494, so we retain the no-drift specification.

Parameter	Estimate	Std. error	z statistic	p-value	95% lower	95% upper
AR(1)	0.242541	0.063408	3.825058	0.000131	0.118263	0.366819
AR(2)	-0.198845	0.063399	-3.136419	0.00171	-0.323104	-0.074586
sigma2	1.323609	0.121081	10.931569	8.14e-28	1.086294	1.560924

Table 2. *Parameter Estimates for ARIMA(2,1,0) without drift*

Table 2 reports the fitted ARIMA(2,1,0) parameters. Both AR coefficients are significant at the 5% level (0.2425 and −0.1988), and the estimated innovation variance is 1.3236. The fitted model is

$$\Delta Y_t = 0.2425\Delta Y_{t-1} - 0.1988\Delta Y_{t-2} + \varepsilon_t$$

Lag	Adjusted df	Q statistic	p-value	Reject at 0.05
6	4	1.058651	0.900776	No
10	8	9.106966	0.333353	No
12	10	11.383613	0.328422	No
18	16	15.752465	0.470367	No
24	22	19.145162	0.636348	No

Table 3. *Adjusted Ljung–Box tests for ARIMA(2,1,0) without drift*

Note: Degrees of freedom are adjusted for two estimated AR parameters.

The adjusted Ljung–Box tests show no significant overall residual autocorrelation at lags 6, 10, 12, 18, or 24 (all $p > 0.05$; Table 3). Although the residual ACF has an isolated spike at lag 10 (0.1528), the corresponding Ljung–Box test is not significant ($p = 0.3334$). The diagnostics therefore support the model's adequacy in capturing short-run linear dependence.

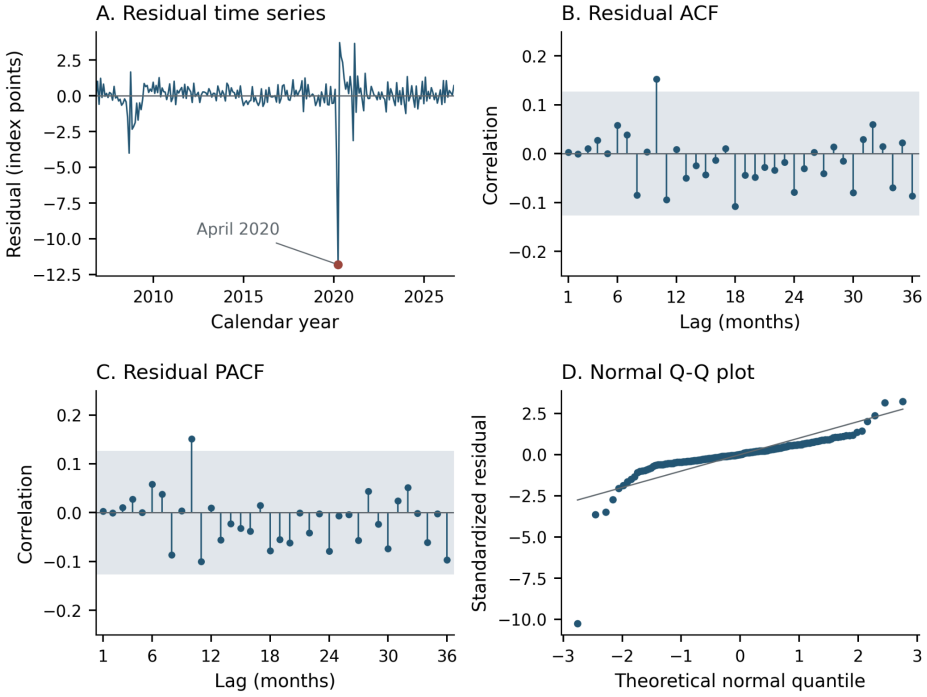


Figure 3. Residual diagnostics for ARIMA(2,1,0) without drift.

239 errors, November 2006-September 2026; initial diffuse error excluded.
ACF/PACF: 95% pointwise reference bands; lags 1-36. Q-Q standardized for display.

Although the Ljung–Box tests show no significant overall residual autocorrelation, Figure 3 reveals important distributional departures. A large negative residual occurs in April 2020 (approximately −11.82 index points), while the normal Q-Q plot indicates substantial deviations from Gaussianity, particularly in the lower tail. These findings raise concerns about Gaussian prediction intervals despite the model's adequate fit to short-run linear dependence.

Forecasting and Prediction Intervals

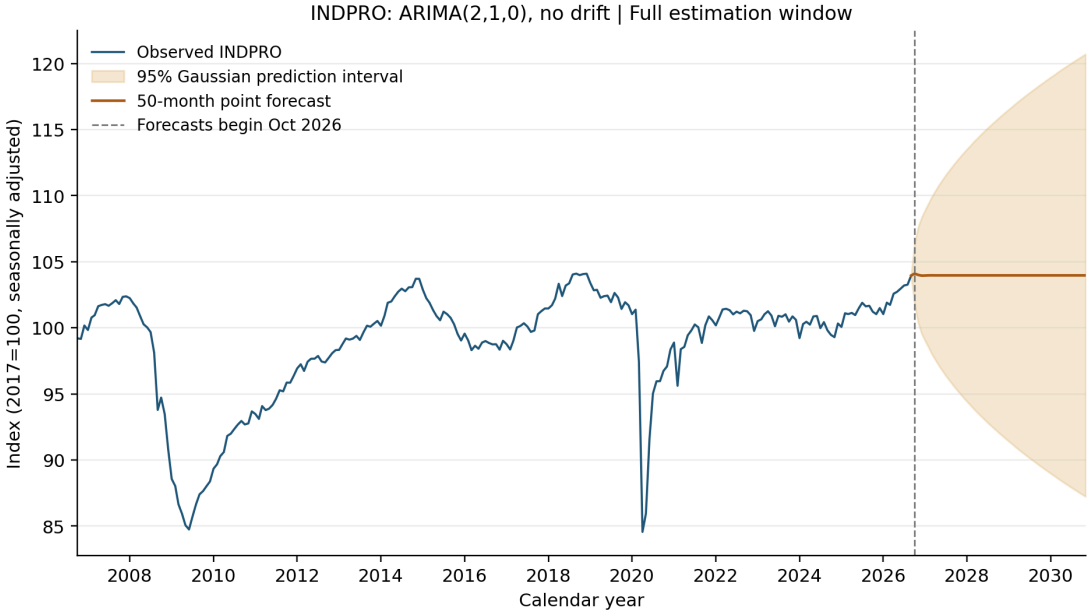


Figure 4. *Fifty-month INDPRO forecasts and 95% Gaussian prediction intervals from ARIMA(2,1,0) without drift*

Figure 4 presents 50-month forecasts from October 2026 through November 2030. The forecasts appear reasonable as a short-run statistical baseline because industrial production is highly persistent and the predicted level remains close to the latest observation. However, the flat long-run forecast reflects the no-drift ARIMA specification rather than an economic expectation of zero growth.

h	Gaussian lower	Gaussian upper	Gaussian width
1	101.8486	106.3584	4.5098
12	95.6765	112.2598	16.5833
24	92.3290	115.6073	23.2783
50	87.2342	120.7022	33.4679

Table 4. *95% prediction interval of ARIMA(2,1,0) without drift*

The 95% prediction intervals are relatively narrow at short horizons but become excessively wide for precise long-term forecasting. At 50 months, the interval spans [87.23, 120.70], covering values near the major historical crisis troughs. This reflects accumulating uncertainty under the integrated model and limits the practical usefulness of long-horizon forecasts.

Bootstrap Sensitivity Analysis and Discussion

The heavy-tailed residuals in Figure 3 motivate a bootstrap assessment of the Gaussian prediction intervals. Holding the fitted ARIMA parameters fixed, we independently resample 239 centered residuals to simulate 20,000 future paths. The 2.5th and 97.5th percentiles provide 95% bootstrap prediction intervals.

Forecast horizon	Gaussian PI width	Bootstrap PI width	Relative difference
1 month	4.510	3.531	−21.7%
12 months	16.583	18.674	+12.6%
24 months	23.278	24.172	+3.8%
50 months	33.468	33.173	−0.9%

Table 5. *Comparison of Gaussian and Bootstrap Prediction Intervals*

Table 5 shows that the bootstrap interval is 21.7% narrower than the Gaussian interval at one month but 12.6% wider at 12 months. The latter suggests greater uncertainty under the empirical innovation distribution, although interval width alone does not establish greater downside risk. By 50 months, the interval widths differ by less than 1%.

These differences demonstrate sensitivity to the innovation distribution, not superior coverage. Holding parameters fixed omits estimation uncertainty, potentially understating long-horizon forecast uncertainty. Independent residual resampling also cannot preserve volatility clustering if present, while extreme observations such as April 2020 may strongly influence the simulated tails. Without out-of-sample validation, the empirical coverage of either interval remains unknown.

Conclusion

ARIMA(2,1,0) without drift provides a parsimonious baseline for U.S. industrial production, achieving the lowest AICc among candidate models and showing no significant overall residual autocorrelation. However, the unit-root specification implies persistent shocks and growing

long-horizon uncertainty. Bootstrap results further show that prediction intervals depend on innovation-distribution assumptions, particularly in the presence of extreme observations. These findings support the model as a descriptive forecasting baseline but do not establish its out-of-sample predictive accuracy.

References

Board of Governors of the Federal Reserve System. (n.d.). *Industrial production: Total index [INDPRO]* [Data set]. Federal Reserve Bank of St. Louis, FRED. Retrieved October 22, 2026, from https://fred.stlouisfed.org/series/INDPRO